



COMSATS - ATD

# Lab Task 3

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## Software Quality Engineering

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# Lab 03 – Understanding and Practicing CMMI Level 1, 2 and 3

## Homework: Multi-Level Scenario Analysis & Process Transformation

### Scenario

**EduTech Solutions** is a software vendor that develops library and administrative software. The company is facing project delays, frequent post-release defects, high employee turnover, and inconsistent code quality across development teams.

The organization currently demonstrates several characteristics of **CMMI Level 1 (Initial)**, where processes are mostly ad hoc, reactive, and dependent on individual employees.

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## Task 1: Defect & Process Identification

The five observed workplace practices are mapped below to the major **CMMI Level 1 symptoms**.

| Issue # | Identified Failure                                                                                        | Level 1 Symptom           | Explanation                                                                                                                                                 |
|---------|-----------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Developers implement client-requested features immediately without checking conflicts with existing code. | <b>Reactive / Chaotic</b> | Developers act immediately instead of following a controlled requirements and change-management process. This can introduce conflicts, defects, and rework. |
| 2       | Project managers estimate timelines based on "gut feeling" without recording historical metrics.          | <b>Chaotic</b>            | Estimates are based on personal judgment rather than documented data, historical performance, or measurement.                                               |

|   |                                                                                                                    |                                           |                                                                                                                                                                            |
|---|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | Each developer uses their own folder structures and naming conventions; knowledge is lost when a developer leaves. | <b>Lack of Standardization</b>            | There are no common coding, naming, or organizational standards. Knowledge remains with individuals instead of being maintained through standardized project practices.    |
| 4 | Subcontractors are hired on short notice without formal quality checks on their deliverables.                      | <b>Chaotic / Reactive</b>                 | Supplier selection and deliverable evaluation are not controlled through a defined process. Problems may only be discovered after the supplier delivers poor-quality work. |
| 5 | Testing consists of a quick manual walkthrough immediately before deployment.                                      | <b>Reactive / Lack of Standardization</b> | Testing is performed at the last moment instead of being planned throughout development. There are also no standardized test cases, checklists, or acceptance criteria.    |

### Summary of Level 1 Problems

The five failures demonstrate the three major characteristics of **CMMI Level 1**:

#### 1. Chaotic:

Work is performed without reliable procedures, measurements, or controls.

#### 2. Reactive:

The organization responds to problems after they occur instead of preventing them.

#### 3. Lack of Standardization:

Different developers or teams follow different methods, resulting in inconsistent quality.

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## Task 2: Level 2 (Managed) Remediation Plan

The objective of CMMI Level 2 is to introduce **project-level management and quality controls** so that processes become **planned, monitored, repeatable, and controlled**.

The five failures can be addressed using the following Level 2 Process Areas:

| Issue # | Identified Failure                                                             | Proposed Level 2 Process Area                       | Concrete QA Mechanism (Checklist, Plan, Log)                                                                                                                                               |
|---------|--------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Direct implementation of client-requested features without checking conflicts. | <b>REQM – Requirements Management</b>               | Establish a <b>Requirements Change Control Form</b> , maintain a <b>Requirements Traceability Matrix (RTM)</b> , and require impact analysis and approval before implementing changes.     |
| 2       | "Gut feeling" project estimations without historical metrics.                  | <b>PP – Project Planning</b>                        | Create a <b>Project Estimation Sheet</b> containing effort, resources, budget, milestones, assumptions, and schedule. Estimates should be documented and reviewed before project approval. |
| 3       | Inconsistent folder structures and naming conventions between developers.      | <b>CM – Configuration Management</b>                | Establish a <b>Configuration Management Plan</b> , standard repository structure, naming conventions, version-control rules, and approved baselines.                                       |
| 4       | Subcontractors hired without formal quality checks.                            | <b>SAM – Supplier Agreement Management</b>          | Use a <b>Supplier Evaluation Checklist</b> , define quality requirements in supplier agreements, and maintain a <b>Supplier Deliverable Review/Acceptance Log</b> .                        |
| 5       | Ad hoc pre-release testing performed by developers.                            | <b>PPQA – Process and Product Quality Assurance</b> | Introduce a <b>QA Testing Checklist</b> , documented test cases, defect logs, acceptance criteria, and a <b>QA release gate</b> before deployment.                                         |

## Detailed Explanation of the Five Level 2 Controls

### 1. REQM – Requirements Management

#### Problem

Developers immediately implement requested features without checking their impact on the existing system.

#### Level 2 Solution

EduTech Solutions should introduce a formal **requirements change-control process**.

### **QA Mechanisms**

- Requirements Change Request Form
- Requirements Traceability Matrix
- Impact Analysis Checklist
- Requirement approval record

### **Process**

**Client Request → Change Request → Impact Analysis → Approval → Development → Testing → Release**

This prevents developers from making uncontrolled changes that may introduce conflicts or defects.

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## **2. PP – Project Planning**

### **Problem**

Project managers estimate schedules based on personal "gut feeling."

### **Level 2 Solution**

Project estimates should be formally documented during project planning.

### **QA Mechanisms**

- Project Planning Document
- Estimation Worksheet
- Resource Allocation Plan
- Schedule/Milestone Plan

For example:

| <b>Activity</b> | <b>Estimated Effort</b> | <b>Resources</b> | <b>Planned Completion</b> |
|-----------------|-------------------------|------------------|---------------------------|
| Requirements    | 40 hours                | 2 members        | Week 1                    |
| Design          | 50 hours                | 2 members        | Week 2                    |

|             |           |           |          |
|-------------|-----------|-----------|----------|
| Development | 160 hours | 4 members | Week 3–5 |
| Testing     | 80 hours  | 2 testers | Week 6   |

This makes project planning more systematic and provides a documented basis for future estimates.

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### 3. CM – Configuration Management

#### Problem

Every developer uses different folder structures and naming conventions.

#### Level 2 Solution

The team should establish a common configuration-management process.

#### QA Mechanisms

- Configuration Management Plan
- Standard folder structure
- Naming convention checklist
- Version-control repository
- Baseline records

#### Example Standard Structure

```
EduTech-Portal/
├── src/
│   ├── components/
│   ├── pages/
│   ├── services/
│   └── utils/
├── tests/
├── docs/
├── config/
└── README.md
```

This reduces confusion and makes the project easier to maintain when a developer leaves.

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## 4. SAM – Supplier Agreement Management

### Problem

Subcontractors are selected quickly without formally checking the quality of their work.

### Level 2 Solution

EduTech should establish formal supplier evaluation and monitoring procedures.

### QA Mechanisms

- Supplier Selection Checklist
- Supplier Quality Requirements
- Contract/Agreement
- Deliverable Acceptance Criteria
- Supplier Evaluation Log

### Example Supplier Checklist

| Criteria                           | Status |
|------------------------------------|--------|
| Required technical skills verified | ✓      |
| Previous work reviewed             | ✓      |
| Quality requirements defined       | ✓      |
| Delivery deadline defined          | ✓      |
| Acceptance criteria defined        | ✓      |
| Security requirements defined      | ✓      |

The supplier's deliverables should be reviewed against these requirements before acceptance.

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## 5. PPQA – Process and Product Quality Assurance

### Problem

Testing only happens through a quick manual walkthrough immediately before deployment.

### Level 2 Solution

EduTech should introduce a formal QA process for reviewing both **processes and products**.

### QA Mechanisms

- Test Case Checklist
- Defect Log
- Code Review Checklist
- Test Execution Record
- Release QA Checklist

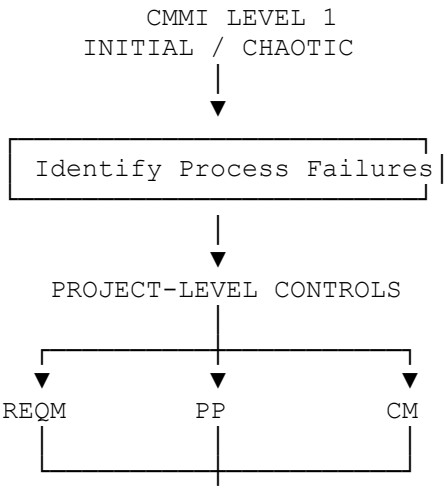
Example Release QA Checklist

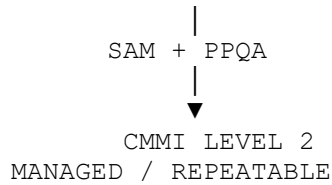
| QA Check                      | Status |
|-------------------------------|--------|
| Requirements verified         | ✓      |
| Test cases executed           | ✓      |
| Critical defects resolved     | ✓      |
| Regression testing completed  | ✓      |
| Code review completed         | ✓      |
| Acceptance criteria satisfied | ✓      |
| QA approval obtained          | ✓      |

Only after the required checks are completed should the software be released.

# Final Transformation

The overall transformation of EduTech Solutions can be represented as:





At **Level 2**, EduTech does not necessarily have one standard process for the entire organization yet. Instead, individual projects have established controls for **requirements, planning, monitoring, configuration, suppliers, measurement, and quality assurance**.

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## Expected Improvement After Level 2

| Existing Problem                    | After Level 2                                  |
|-------------------------------------|------------------------------------------------|
| Uncontrolled feature implementation | Controlled requirements and change management  |
| Gut-feeling estimation              | Documented project planning and estimation     |
| Different folder structures         | Controlled configuration standards             |
| Unchecked subcontractors            | Formal supplier evaluation                     |
| Last-minute testing                 | Planned QA and documented testing              |
| Knowledge dependent on individuals  | Project documentation and controlled artifacts |
| Frequent post-release defects       | Earlier detection and formal release checks    |
| Unpredictable project execution     | More repeatable project management             |

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## Conclusion

EduTech Solutions currently operates with many characteristics of **CMMI Level 1**, including chaotic practices, reactive problem-solving, and lack of standardization. The five observed failures demonstrate that important activities such as requirements management, estimation, configuration management, supplier management, and testing are not formally controlled.

To move toward **CMMI Level 2 (Managed)**, the company should introduce project-level controls through **REQM, PP, CM, SAM, and PPQA**. Concrete QA mechanisms such as **change-control forms, traceability matrices, planning documents, configuration plans, supplier checklists, test checklists, and defect logs** make these processes repeatable and measurable.

Therefore, the transition from **Level 1 → Level 2** is essentially a transition from "**individual heroics and ad hoc work**" to "**planned, controlled, and repeatable project processes**."